

BASIC NETWORK CONCEPTS

Complete Study Notes

Topics Covered

1. Importance of OSI Model
2. OSI Model & TCP/IP Model
3. IP Address & Classes
4. Subnet Mask
5. Default Gateway
6. IP Subnetting
7. FLSM / VLSM / CIDR

1. Importance of the OSI Model

The Open Systems Interconnection (OSI) model is a conceptual framework developed by the International Organization for Standardization (ISO) in 1984. It standardizes how different network systems communicate with each other, regardless of their internal structure or technology.

1.1 What is the OSI Model?

- A 7-layer reference model that describes how data travels from one computer to another over a network
- Stands for Open Systems Interconnection
- Published by ISO (International Organization for Standardization) in 1984
- It is a conceptual model — not a protocol itself, but a framework for understanding protocols

1.2 Why is the OSI Model Important?

- **Standardization:** Provides a universal language for networking, allowing products from different vendors to work together
- **Troubleshooting:** Helps network engineers isolate problems at a specific layer (e.g., physical cable vs. software issue)
- **Modular Design:** Each layer has a specific function; changes in one layer don't affect others
- **Vendor Independence:** Devices from different manufacturers can communicate because they follow the same layered standards
- **Protocol Development:** Guides the creation of new networking protocols in a structured way
- **Education:** Provides a clear framework for teaching and learning network concepts

 **Note:** Remember OSI layers using the mnemonic: 'All People Seem To Need Data Processing' (Application → Presentation → Session → Transport → Network → Data Link → Physical)

1.3 OSI Layer Functions Overview

Layer #	Layer Name	Protocols / Devices	Key Function
7	Application	HTTP, FTP, SMTP, DNS, Telnet	User interface to network services
6	Presentation	SSL/TLS, JPEG, MPEG, ASCII	Data translation, encryption, compression
5	Session	NetBIOS, RPC, PPTP	Establishes, manages, terminates sessions
4	Transport	TCP, UDP	End-to-end communication, error recovery
3	Network	IP, ICMP, OSPF, BGP, RIP	Logical addressing and routing
2	Data Link	Ethernet, Wi-Fi, PPP, ARP, MAC	Physical addressing, error detection

1	Physical	Cables, Hubs, Repeaters, NIC	Raw bit transmission over medium
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2. OSI Model & TCP/IP Model

While the OSI model is a theoretical framework, the TCP/IP model (also called the Internet Model or DoD model) is the practical implementation used in real-world networking and the Internet.

2.1 TCP/IP Model Layers vs OSI Layers

OSI Layer	TCP/IP Layer	Protocols	Description
7,6,5	Application	HTTP, FTP, SMTP, DNS, SNMP	Provides network services to applications
4	Transport	TCP, UDP	End-to-end data transfer and reliability
3	Internet	IP, ICMP, ARP, IGMP	Logical addressing and packet routing
2,1	Network Access	Ethernet, Wi-Fi, PPP	Physical transmission of data frames

2.2 Key Differences: OSI vs TCP/IP

- **Number of Layers:** OSI has 7 layers; TCP/IP has 4 layers
- **Development:** OSI is a conceptual model; TCP/IP is a practical, implemented protocol suite
- **Usage:** OSI is used for teaching and troubleshooting; TCP/IP is used for actual Internet communication
- **Session & Presentation:** OSI separates these; TCP/IP combines them into the Application layer
- **Physical Layer:** OSI has a separate Physical layer; TCP/IP includes it in the Network Access layer

2.3 Data Encapsulation

As data travels down the OSI layers, each layer adds its own header (and sometimes trailer). This is called encapsulation:

- **Application Layer (7):** Data
- **Transport Layer (4):** Segment (adds TCP/UDP header with port numbers)
- **Network Layer (3):** Packet (adds IP header with source/destination IP)
- **Data Link Layer (2):** Frame (adds MAC addresses + error check)
- **Physical Layer (1):** Bits (0s and 1s transmitted over medium)

 **Note:** TCP/IP uses 3-way handshake (SYN → SYN-ACK → ACK) to establish a reliable connection before data transfer.

3. IP Address & Classes

An IP (Internet Protocol) address is a unique numerical label assigned to each device on a network. It serves two purposes: host identification and location addressing.

3.1 IPv4 Address Structure

- 32 bits long, written as 4 octets separated by dots (dotted decimal notation)
- Each octet is 8 bits and ranges from 0 to 255
- Example: 192.168.10.100

192	.	168	.	10	.	100	↑	↑	↑	↑
1st Octet 2nd Octet 3rd Octet 4th Octet (Each = 8 bits = 1 Byte)										

3.2 IP Address Classes

Class	1st Octet Range	Default Mask	Network Bits	Host Bits	Example / Use Case
A	1 – 126	255.0.0.0 /8	8	24	10.0.0.0 – Large enterprises/ISPs
B	128 – 191	255.255.0.0 /16	16	16	172.16.0.0 – Medium organizations
C	192 – 223	255.255.255.0 /24	24	8	192.168.1.0 – Small networks/LANs
D	224 – 239	N/A (Multicast)	—	—	Multicasting (streaming, routing protocols)
E	240 – 255	N/A (Reserved)	—	—	Research and experimental use only

 **Note:** 127.0.0.1 is the loopback address (localhost) — used to test the TCP/IP stack on a device. The range 127.0.0.0/8 is reserved for loopback.

3.3 Private vs Public IP Addresses

Private IP addresses are not routable on the Internet. They are used within local networks (LANs):

Class	Private Range	CIDR	# of Addresses
A	10.0.0.0 – 10.255.255.255	/8	16,777,216
B	172.16.0.0 – 172.31.255.255	/12	1,048,576
C	192.168.0.0 – 192.168.255.255	/16	65,536

- **Public IPs:** Assigned by ISPs; routable on the Internet
- **Private IPs:** Used inside LANs; requires NAT to reach the Internet

3.4 Special IP Addresses

- **0.0.0.0:** Represents 'this network' or unspecified address
- **255.255.255.255:** Limited broadcast — sent to all hosts on local network
- **127.0.0.1:** Loopback address — tests local TCP/IP stack
- **169.254.x.x (APIPA):** Auto-assigned when DHCP fails — Automatic Private IP Addressing

4. Subnet Mask

A subnet mask is a 32-bit number that divides an IP address into the Network portion and the Host portion. It helps routers and devices determine which part of the IP address identifies the network and which part identifies the host.

4.1 How Subnet Masks Work

- Binary 1s in the mask represent the Network portion
- Binary 0s in the mask represent the Host portion
- Devices perform a bitwise AND operation between the IP and mask to find the network address

Example: IP = 192.168.1.100 | Mask = 255.255.255.0

Network Address: 192.168.1.0 | Broadcast: 192.168.1.255 | Usable Hosts:
192.168.1.1 – 192.168.1.254

4.2 Common Subnet Masks

CIDR	Subnet Mask	Binary Representation	Usable Hosts	Subnets from /24
/24	255.255.255.0	11111111.11111111.1111 1111.00000000	254	1
/25	255.255.255.128	11111111.11111111.1111 1111.10000000	126	2
/26	255.255.255.192	11111111.11111111.1111 1111.11000000	62	4
/27	255.255.255.224	11111111.11111111.1111 1111.11100000	30	8
/28	255.255.255.240	11111111.11111111.1111 1111.11110000	14	16
/29	255.255.255.248	11111111.11111111.1111 1111.11111000	6	32
/30	255.255.255.252	11111111.11111111.1111 1111.11111100	2	64

4.3 Calculating Usable Hosts

Usable Hosts = $2^n - 2$ (where n = number of host bits)

The 2 subtracted accounts for: Network Address (all 0s) and Broadcast Address (all 1s)

5. Default Gateway

The Default Gateway is the IP address of the router interface that acts as the exit point for devices on a local network to communicate with devices on other networks (including the Internet).

5.1 What Does a Default Gateway Do?

- Acts as a 'door' between the local network and external networks
- When a host wants to communicate with a device outside its subnet, it sends traffic to the default gateway
- The gateway/router then forwards packets toward the destination network
- Without a default gateway, a host can only communicate within its own subnet

5.2 How the Default Gateway is Used

- **Step 1:** Host checks if destination IP is in the same subnet using subnet mask
- **Step 2:** If SAME subnet → communicate directly (ARP is used to find MAC address)
- **Step 3:** If DIFFERENT subnet → send packet to default gateway's IP address
- **Step 4:** Router receives packet and routes it toward the destination

5.3 Typical Home/Office Network

Device	IP Address	Role
Router (Default Gateway)	192.168.1.1	Connects LAN to Internet
PC / Workstation	192.168.1.100	End-user device on LAN
Printer / IoT Device	192.168.1.150	Local network device

 **Note:** Always configure the default gateway to the router's LAN IP address. If misconfigured, devices can communicate locally but cannot reach the Internet.

6. IP Subnetting

Subnetting is the process of dividing a large network into smaller, more manageable sub-networks (subnets). It improves performance, security, and efficient use of IP address space.

6.1 Why Subnet?

- **Reduce Broadcast Domains:** Fewer hosts per subnet = fewer broadcast packets
- **Improve Security:** Separate departments (HR, Finance) into isolated subnets
- **Efficient IP Use:** Allocate only the addresses needed for each segment
- **Better Performance:** Less congestion; traffic stays local within subnets
- **Logical Organization:** Group devices by location, function, or department

6.2 Subnetting Formulas

What to Calculate	Formula	Example (/26 from /24)
Number of Subnets	2^s	$2^2 = 4$ subnets (2 subnet bits)
Hosts per Subnet	$2^h - 2$	$2^6 - 2 = 62$ usable hosts
Block Size (Increment)	$256 - \text{mask octet}$	$256 - 192 = 64$

6.3 Step-by-Step Subnetting Example

Given: 192.168.1.0/26 → Subnet Mask: 255.255.255.192

- **Block Size:** $256 - 192 = 64$
- **Subnet 1:** 192.168.1.0/26 → Hosts: .1 to .62 → Broadcast: .63
- **Subnet 2:** 192.168.1.64/26 → Hosts: .65 to .126 → Broadcast: .127
- **Subnet 3:** 192.168.1.128/26 → Hosts: .129 to .190 → Broadcast: .191
- **Subnet 4:** 192.168.1.192/26 → Hosts: .193 to .254 → Broadcast: .255

 **Note:** Quick Trick: Memorize powers of 2 — 2, 4, 8, 16, 32, 64, 128, 256. The block size is always a power of 2.

7. FLSM / VLSM / CIDR

7.1 FLSM – Fixed Length Subnet Masking

FLSM divides a network into subnets of equal size. Every subnet has the same subnet mask, resulting in the same number of hosts per subnet.

- **Characteristic:** All subnets use the same prefix length
- **Advantage:** Simple to configure and manage
- **Disadvantage:** Wastes IP addresses when subnets vary in size
- **Used With:** Older routing protocols like RIPv1 (classful)

FLSM Example — Dividing 192.168.1.0/24 into 4 equal subnets (/26):

- **Subnet 1:** 192.168.1.0/26 (62 hosts)
- **Subnet 2:** 192.168.1.64/26 (62 hosts)
- **Subnet 3:** 192.168.1.128/26 (62 hosts)
- **Subnet 4:** 192.168.1.192/26 (62 hosts)

7.2 VLSM – Variable Length Subnet Masking

VLSM allows different subnets within the same network to have different subnet masks. Each subnet is sized according to the actual number of hosts required.

- **Characteristic:** Different subnets use different prefix lengths
- **Advantage:** Very efficient — minimizes wasted addresses
- **Disadvantage:** More complex to design and document
- **Used With:** Modern protocols: RIPv2, OSPF, EIGRP, BGP

VLSM Example — Allocating from 192.168.1.0/24:

- **Dept A (60 hosts):** 192.168.1.0/26 → mask /26 (62 usable)
- **Dept B (30 hosts):** 192.168.1.64/27 → mask /27 (30 usable)
- **Dept C (12 hosts):** 192.168.1.96/28 → mask /28 (14 usable)
- **Point-to-Point Link:** 192.168.1.112/30 → mask /30 (2 usable)

7.3 FLSM vs VLSM Comparison

Feature	FLSM (Fixed Length)	VLSM (Variable Length)
Subnet Size	All subnets are the same size	Each subnet can be a different size
IP Utilization	Wasteful – many unused addresses	Efficient – minimal wastage
Complexity	Simple to implement and manage	More complex to plan and implement
Scalability	Poor – does not scale well	Excellent – highly scalable
RIP Support	RIPv1 (classful routing)	RIPv2, OSPF, EIGRP (classless)

CIDR Support	No	Yes
Best For	Small, uniform networks	Large, enterprise-scale networks

7.4 CIDR – Classless Inter-Domain Routing

CIDR (pronounced 'cider') was introduced in 1993 (RFC 1519) to replace the old classful IP addressing system. It allows IP addresses to be allocated in variable-size blocks, independent of the old Class A/B/C structure.

- **Notation:** IP address followed by /prefix-length e.g., 192.168.1.0/24
- **Purpose:** Reduces routing table size and slows IPv4 exhaustion
- **Supernetting:** CIDR allows combining multiple networks into one summary route
- **Route Aggregation:** E.g., 192.168.0.0/22 summarizes .0, .1, .2, .3 /24 networks

7.5 Common CIDR Blocks

CIDR Block	Subnet Mask	Total Hosts	Example Network
/8	255.0.0.0	16,777,216	10.0.0.0/8 (Class A Private)
/16	255.255.0.0	65,536	172.16.0.0/16 (Class B Private)
/24	255.255.255.0	256	192.168.1.0/24 (Small LAN)
/28	255.255.255.240	16	192.168.1.16/28 (Small Dept)
/30	255.255.255.252	4	Point-to-Point links (2 hosts)
/32	255.255.255.255	1	Host route / loopback address

7.6 CIDR Route Aggregation Example

Aggregating four /24 networks into a single /22 summary route:

- **Network 1:** 192.168.0.0/24
- **Network 2:** 192.168.1.0/24
- **Network 3:** 192.168.2.0/24
- **Network 4:** 192.168.3.0/24

Summary Route: 192.168.0.0/22 → covers all 4 networks above

 **Note:** CIDR is classless — it ignores the old Class A/B/C boundaries. Any prefix length from /1 to /32 is valid.

8. FLSM & VLSM – Numerical Examples

This section provides complete step-by-step worked examples for both FLSM and VLSM subnetting. Work through each example carefully to master subnet calculations.

8.1 FLSM Numerical Example 1

Problem: Divide 192.168.10.0/24 into 4 equal subnets

Step 1 – Determine how many subnet bits are needed

- We need 4 subnets $\rightarrow 2^s \geq 4 \rightarrow s = 2$ bits
- New prefix = $/24 + 2 = /26$
- New Subnet Mask = 255.255.255.192

Step 2 – Calculate block size

- Block Size = $256 - 192 = 64$
- Each subnet spans 64 addresses

Step 3 – Calculate hosts per subnet

- Host bits = $32 - 26 = 6$ bits
- Usable Hosts = $2^6 - 2 = 64 - 2 = 62$ hosts per subnet

Step 4 – List all subnets

Subnet	Network Address	First Host	Last Host	Broadcast
1	192.168.10.0/26	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64/26	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128/26	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192/26	192.168.10.193	192.168.10.254	192.168.10.255

 **Note:** FLSM key: All 4 subnets have the SAME mask (/26) and SAME host count (62). Block size (64) is the increment between Network Addresses.

8.2 FLSM Numerical Example 2

Problem: Divide 172.16.0.0/16 into 8 equal subnets

Step 1 – Subnet bits needed

- Need 8 subnets $\rightarrow 2^s \geq 8 \rightarrow s = 3$ bits
- New prefix = $/16 + 3 = /19$
- Subnet Mask = 255.255.224.0

Step 2 – Block size

- Block size in 3rd octet = $256 - 224 = 32$
- Each subnet spans 32 in the 3rd octet (8192 total addresses)

Step 3 – Hosts per subnet

- Host bits = $32 - 19 = 13$ bits
- Usable Hosts = $2^{13} - 2 = 8192 - 2 = 8190$ hosts per subnet

Step 4 – All subnets

Subnet	Network Address	First Host	Last Host	Broadcast
1	172.16.0.0/19	172.16.0.1	172.16.31.254	172.16.31.255
2	172.16.32.0/19	172.16.32.1	172.16.63.254	172.16.63.255
3	172.16.64.0/19	172.16.64.1	172.16.95.254	172.16.95.255
4	172.16.96.0/19	172.16.96.1	172.16.127.254	172.16.127.255
5	172.16.128.0/19	172.16.128.1	172.16.159.254	172.16.159.255
6	172.16.160.0/19	172.16.160.1	172.16.191.254	172.16.191.255
7	172.16.192.0/19	172.16.192.1	172.16.223.254	172.16.223.255
8	172.16.224.0/19	172.16.224.1	172.16.255.254	172.16.255.255

8.3 VLSM Numerical Example 1

Problem: Allocate from 192.168.1.0/24 for depts needing: 50, 25, 10, and 2 hosts

VLSM Rule: Always allocate the LARGEST requirement first to avoid fragmentation.

Step 1 – Sort requirements largest to smallest

Priority	Department	Hosts Needed	Prefix /n	Usable Hosts
1st	Sales	50	/26	62 (2^6-2)
2nd	HR	25	/27	30 (2^5-2)
3rd	IT	10	/28	14 (2^4-2)
4th	P-to-P Link	2	/30	2 (2^2-2)

Step 2 – Allocate subnets sequentially

Dept	Prefix	Network Address	Host Range	Broadcast
Sales	/26	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
HR	/27	192.168.1.64/27	192.168.1.65 – 192.168.1.94	192.168.1.95
IT	/28	192.168.1.96/28	192.168.1.97 – 192.168.1.110	192.168.1.111
P-to-P	/30	192.168.1.112/30	192.168.1.113 – 192.168.1.114	192.168.1.115
Unused	—	192.168.1.116 onward	140 addresses remain unused	192.168.1.255

 **Note:** VLSM efficiency: Only 116 of 256 addresses allocated. FLSM using /26 for all 4 would allocate all 256 and still waste 161 addresses on small subnets.

8.4 VLSM Numerical Example 2

Problem: Allocate from 10.0.0.0/24 for: Finance(100), Admin(50), Ops(20), Mgmt(5), WAN(2)

Step 1 – Choose prefix for each requirement

Department	Hosts Needed	Prefix	Block Size	Calculation / Justification
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Finance	100	/25	128	$2^7 - 2 = 126 \geq 100$ (host bits = 7)
Admin	50	/26	64	$2^6 - 2 = 62 \geq 50$ (host bits = 6)
Ops	20	/27	32	$2^5 - 2 = 30 \geq 20$ (host bits = 5)
Mgmt	5	/29	8	$2^3 - 2 = 6 \geq 5$ (host bits = 3)
WAN Link	2	/30	4	$2^2 - 2 = 2 \geq 2$ (host bits = 2)

Step 2 – Allocate sequentially (largest subnet first)

Dept	Prefix	Network Address	Host Range	Broadcast
Finance	/25	10.0.0.0/25	10.0.0.1 – 10.0.0.126	10.0.0.127
Admin	/26	10.0.0.128/26	10.0.0.129 – 10.0.0.190	10.0.0.191
Ops	/27	10.0.0.192/27	10.0.0.193 – 10.0.0.222	10.0.0.223
Mgmt	/29	10.0.0.224/29	10.0.0.225 – 10.0.0.230	10.0.0.231
WAN Link	/30	10.0.0.232/30	10.0.0.233 – 10.0.0.234	10.0.0.235
Unused	—	10.0.0.236 onwards	20 addresses remaining free	10.0.0.255

 **Note:** Notice /29 was chosen for Mgmt (5 hosts needed, 6 usable) rather than /28 (14 usable). Always choose the smallest mask that fits — never over-allocate.

8.5 FLSM vs VLSM – Address Waste Comparison

Same scenario: Depts needing 50, 25, 10, 2 hosts from 192.168.5.0/24

Dept (Need)	Hosts Needed	Method	Mask Used	Usable Hosts	Wasted IPs
Sales (50)	50	FLSM	/26 – 255.255.255.192	62	12
HR (25)	25	FLSM	/26 – 255.255.255.192	62	37
IT (10)	10	FLSM	/26 – 255.255.255.192	62	52
Link (2)	2	FLSM	/26 – 255.255.255.192	62	60
FLSM TOTAL	87	—	4 × /26 = 256 addresses	248	161 wasted!
Sales (50)	50	VLSM	/26 – 255.255.255.192	62	12
HR (25)	25	VLSM	/27 – 255.255.255.224	30	5
IT (10)	10	VLSM	/28 – 255.255.255.240	14	4
Link (2)	2	VLSM	/30 – 255.255.255.252	2	0
VLSM TOTAL	87	—	Variable masks	108	21 wasted only

VLSM saves 140 IP addresses compared to FLSM for the exact same requirements!

8.6 Subnetting Quick Formula Reference Card

Formula	Expression	Example
Number of Subnets	2^s (s = subnet bits borrowed)	s=3 → $2^3 = 8$ subnets
Hosts per Subnet	$2^h - 2$ (h = host bits remaining)	h=6 → $2^6 - 2 = 62$ hosts
Block Size	256 – last non-zero mask octet	mask .192 → $256 - 192 = 64$
Next Network Addr	Current Network + Block Size	$10.0.0.0 + 64 = 10.0.0.64$
Broadcast Address	Next Network – 1	$10.0.0.64 - 1 = 10.0.0.63$
First Usable Host	Network Address + 1	$10.0.0.0 + 1 = 10.0.0.1$
Last Usable Host	Broadcast – 1	$10.0.0.63 - 1 = 10.0.0.62$

Total Addresses	2^h (includes network & broadcast)	$h=6 \rightarrow 2^6 = 64$ total
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 **Note:** VLSM golden rule: Sort requirements from *LARGEST* to *SMALLEST*. Allocate in that order. Always pick the smallest subnet mask that satisfies the host count requirement.

Quick Reference Summary

Topic	Key Points
OSI Model	7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application. Used for troubleshooting and education.
TCP/IP Model	4 layers: Network Access, Internet, Transport, Application. Practical model used for Internet communication.
IP Address	32-bit number in 4 octets. Classes A (1-126), B (128-191), C (192-223), D (Multicast), E (Research).
Subnet Mask	Divides IP into Network + Host parts. Written as dotted decimal (255.x.x.x) or CIDR prefix (/x).
Default Gateway	Router IP for inter-network communication. Without it, devices can't communicate outside their subnet.
Subnetting	Divides networks into smaller subnets. Reduces broadcasts, improves security and performance.
FLSM	Fixed subnets — all same size. Simple but wasteful. Used with classful protocols like RIPv1.
VLSM	Variable subnets — sized by need. Efficient. Used with OSPF, EIGRP, RIPv2.
CIDR	Classless notation (e.g., /24). Enables route aggregation and efficient IP allocation.

— End of Notes —